Does Depth-Scaled Gaussian Initialization Make Quantum Neural Networks Trainable? A Reproduction and Its Extension to Classification and Noisy Hardware

Kazi Muktadir Ahmed, *Student ID: 202584575*

***Abstract*—[Write: About 90 words. (1) Deep quantum neural networks (QNNs) suffer barren plateaus: gradients vanish exponentially with the number of qubits. (2) Zhang et al. (NeurIPS 2022) proved that a depth-scaled Gaussian initialization keeps gradients polynomially large, but tested it only on physics and chemistry tasks. (3) This project reproduces their result and tests the initialization on a supervised QNN classifier, both noiseless and in a noisy hardware simulation. (4) Optional headline: at 12 qubits the Gaussian initialization gives a 147× larger initial gradient than uniform initialization.]**

***Index Terms*—Quantum machine learning, quantum neural networks, parameter initialization, vanishing gradients, barren plateaus, noisy simulation.**

I. Problem

[Write: 2–3 sentences, framed as an ML problem (assume ML knowledge; no need to explain vanishing gradients or initialization schemes). Weight initialization decides whether a deep model is trainable (cf. Xavier/He for classical networks). For QNNs the failure mode is the barren plateau: with uniformly random initial parameters the gradient variance shrinks exponentially in the number of qubits [1], and QML classifiers inherit it from deep circuits and from the data encoding [2]. Number for scale: gradient variance $\sim 2^{-N}$, so at $N = 15$ a typical gradient ≈ 0.0055 needs ∼33,000 measurements to resolve.]

II. Quantum Computing Background

[Write: Introduce only what the proposal uses; cite so readers can follow up (textbook [3], VQA review [4]):

- *qubit / state*: N qubits → a vector of 2^N complex amplitudes ($2^{15} = 32{,}768$; why 15 qubits fit on a laptop);
- *gates*: rotations $R_X(\theta) = e^{-i\theta X}$, $R_Y(\theta) = e^{-i\theta Y}$ (Pauli X, Y, Z; the angles are the trainable "weights"), and the entangling CZ gate;
- *QNN*: data encoding (each input feature sets one rotation angle) → layers of trainable rotations and CZ gates → output = expectation value of Z on one qubit, estimated from repeated measurements ("shots"), error $\sim 1/\sqrt{\text{shots}}$ [3], [4];
- *gradients*: the parameter-shift rule, two circuit runs per parameter [5];
- *Hamiltonian / ground energy* (for the reproduction only): minimizing $\langle H \rangle$ finds the lowest energy; each term acts on S qubits ("locality");
- *device noise*: gate errors, decoherence (T_1/T_2), readout error [3]; IBM "fake backends" = noise models calibrated from real chips, run in Qiskit [6].

]

III. Method Under Study

[Write: Zhang et al. [7], NeurIPS 2022. Cover:

- the rule $\theta \sim \mathcal{N}(0, \gamma^2)$, $\gamma^2 = 1/(4S(L+2))$: variance shrinks with depth L, like Xavier/He shrink it with width; for the classifier ($S = 1$, $L = 18$) $\gamma^2 = 1/80$, vs. uniform $[0, 2\pi]$ variance 3.29;
- Theorem 4.1: $\mathbb{E}\|\nabla f\|^2 \geq L/(S^S(L+2)^{S+1})$, i.e. polynomial, not exponential, decay;
- their evidence: a 15-qubit Heisenberg-chain energy minimization and an LiH molecule; the paper names QML classification as a target use (Sec. 3) but does not test it (no need to explain GD/Adam).

]

IV. Plan

[Write: Three studies, in priority order:

- **S1 Reproduction (validation):** the paper's 15-qubit Heisenberg runs (Gaussian/uniform/zero init × {GD, Adam} × {exact, noisy gradients} × 5 seeds, 100 iterations) in PennyLane;
- **S2 QNN classifier (main ML study):** handwritten digits 3 vs 8 (scikit-learn), PCA to N features, angle encoding, 10 layers, MSE loss on labels ± 1; gradient vs. qubits $N = 4$–14; training at $N = 12$ (240 parameters) with Gaussian/uniform/zero init, Adam, 5 seeds; test accuracy and training curves;
- **S3 Noisy hardware simulation (deployment):** Qiskit Aer with IBM device noise models (5 and 7 qubits), noiseless vs. noisy gradients and training;
- timeline: results by Nov 20, slides Nov 23, report Dec 7.

]

V. Contribution

[Write: Lead with S2+S3: the first test, to our knowledge, of Zhang et al.'s variance rule on a *deep* QNN classifier where barren plateaus actually occur, and under calibrated device noise. Contrast: a 2026 benchmark of initializations for QNN

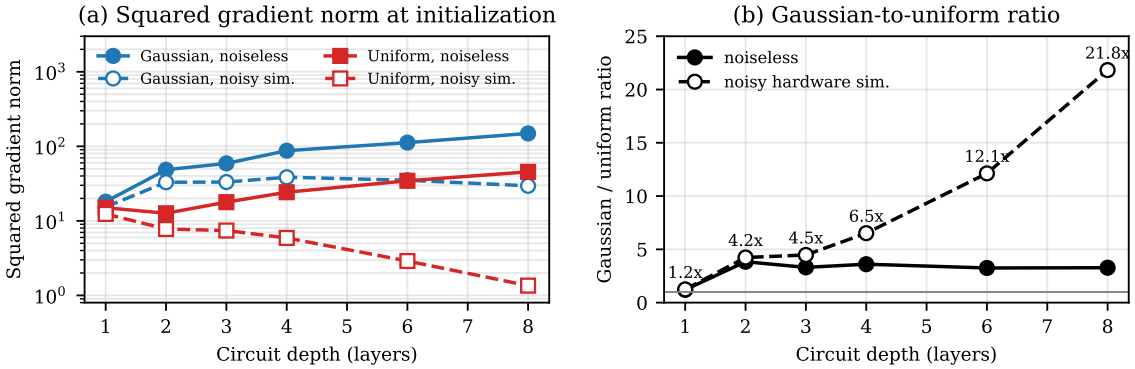


Fig. 1. Noisy hardware simulation (Qiskit Aer, density matrix, FakeManilaV2 noise model), $N = 5$, mean over 8 random draws. (a) Squared gradient norm at initialization vs. depth. (b) Gaussian/uniform ratio, noiseless vs. noisy (2-qubit gate count after transpilation grows from 17 at 1 block to 154 at 8 blocks).

TABLE I
QNN CLASSIFIER (DIGITS 3 VS. 8, 10 LAYERS, MSE LOSS): MEAN SQUARED GRADIENT NORM AT INITIALIZATION OVER 5 RANDOM INITIALIZATIONS AND 16 TRAINING IMAGES.

Qubits	Params	Gaussian	Uniform	Zero	G/U
4	80	5.68	3.98	1.03	1.4
6	120	10.19	1.35	1.06	7.5
8	160	1.27	0.45	1.00	2.8
10	200	6.68	0.12	1.07	55
12	240	8.93	0.061	1.17	147

image classifiers [8] used two-layer circuits, where it found no severe barren plateaus, and no hardware noise; the original paper models noise only as Gaussian noise on gradients; noise itself causes barren plateaus [9]; the closest noise study uses idealized depolarizing noise and a different small-angle start [10]. Then: an independent reproduction (S1).]

VI. Preliminary Work

[Write: 2–4 sentences on Table I and Fig. 1. Facts: the classifier shows a barren plateau: with uniform init the squared gradient norm falls $65\times$ from 4 to 12 qubits ($3.98 \to 0.061$), while Gaussian init stays at 1–10 ($147\times$ larger at 12 qubits); a short 8-qubit training run (60 Adam steps) reaches $\approx$0.90 test accuracy for all inits, with lower test MSE for Gaussian (0.27 vs. 0.46 uniform), so the training difference must be tested at 10–12 qubits. Reproduction check: a GD run gives $f = -19.2$ (it. 10) and -22.8 (it. 100) vs. ≈ -19, -22.8 in the paper's Fig. 2(a). Noisy simulation (Fig. 1): Gaussian/uniform gradient ratio grows $4.2\times \to 21.8\times$ from 2 to 8 layers.]

VII. Experiment Plan

Hypothesis. [Write: e.g. on a 12-qubit QNN classifier, where uniform initialization has vanishing gradients, Gaussian initialization reaches a target test accuracy in fewer training steps, and keeps its advantage in the noisy hardware simulation.]
Proxy. [Write: Statistical: test accuracy, training loss vs. steps, steps to reach a target accuracy, $\mathbb{E}\|\nabla f\|^2$ vs. qubits. Hardware: wall-clock per training step; circuit evaluations per gradient (parameter shift: $2 \times 240 = 480$ per image at 12 qubits); transpiled 2-qubit gate count.]
Protocol. [Write: S2: $N = 12$, 240 parameters, 3 inits $\times$ 5 seeds, Adam (lr 0.01), 100 steps, batch 16, fixed train/test split; $N = 4$–14 gradient sweep. S1: the paper's settings. S3: 5 and 7 qubits, noise on/off, 3 seeds.]
Expected results. [Write: Gaussian $>$ uniform in training speed at $N \geq 10$; zero init competitive on the classifier (the encoded input moves it off the flat point) but not on the Heisenberg task; the advantage persists or widens under noise. Negative results are acceptable.]

References

[1] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, p. 4812, 2018.
[2] S. Thanasilp, S. Wang, N. A. Nghiem, P. J. Coles, and M. Cerezo, "Subtleties in the trainability of quantum machine learning models," *Quantum Machine Intelligence*, 2023.
[3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. Cambridge, U.K.: Cambridge University Press, 2010.
[4] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, pp. 625–644, 2021.
[5] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Physical Review A*, vol. 99, p. 032331, 2019.
[6] A. Javadi-Abhari *et al.*, "Quantum computing with Qiskit," arXiv:2405.08810, 2024.
[7] K. Zhang, L. Liu, M.-H. Hsieh, and D. Tao, "Escaping from the barren plateau via Gaussian initializations in deep variational quantum circuits," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.
[8] M. Rahman *et al.*, "Benchmarking barren plateau mitigation strategies in quantum neural networks on standard and medical image datasets," *Journal of Imaging*, vol. 12, no. 7, p. 275, 2026.
[9] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, "Noise-induced barren plateaus in variational quantum algorithms," *Nature Communications*, vol. 12, p. 6961, 2021.
[10] L. Shen, F. Jin, and B. Guo, "Initialization-controlled fault sensitivity and force-balance plateaus in noisy variational quantum optimization," arXiv:2611.04198, 2026.